

1. Reflection

Our group's journey through the design process has been insightful, offering many lessons that have grown our understanding of engineering design. During brainstorming, we realized the importance of having a diverse team with members who not only brought many perspectives but also created a setting that allowed for respectful collaboration. This dynamic within the design process encouraged diverse concepts, ultimately guiding us to the application of a wheel design to a walker. Moreover, our group's commitment to frequent and inclusive brainstorming sessions aided in our decision-making process. Through active participation from all team members, we ensured that a wide range of ideas was considered with lots of thought poured into each one, building unity in our team for our project's direction. Additionally, being open-minded towards feedback, particularly from our instructor and GSI's, helped us refine our ideas and guide our project towards success. This ultimately manifested in both our elevator pitch and our final design, elevating the quality of our work.

The design process showed us how our project's progression depended on the frequent collaboration with each other. Without constant discussion and communication of our ideas, we wouldn't have been able to fine tune our designs and the details that made our project roll. Similarly, without the invaluable feedback and guidance we received along the way, we would have likely struggled to navigate the complexities of the design process and achieve our desired objectives. Initially, our focus was overly broad, aiming to create a walker with numerous complex mechanisms and functionalities. We failed to consider our developed knowledge of manufacturing and instead dove headfirst into the project with too much ambition and naivete, which became overwhelming as we progressed. However, with the professor's guidance, we simplified our focus to what we wanted to highlight, the wheel. This guidance brought relief and allowed us to dedicate our time to creating the best proof-of-concept design for our wheel and supports possible. This simplicity significantly influenced our manufacturing decisions. Initially, we planned to use metal pipes and weld them together for the walker's frame and to incorporate various complex bearings and belt-driven systems to enhance mobility. However, focusing on the wheel prompted us to opt for simpler methods like laser-cutting and using a handsaw to craft a functional walker body. We also utilized 3D printing to enable rapid prototyping and continue applying improvements to the wheel design. Correcting our focus enabled us to spend more time optimizing each manufacturing process, avoiding suboptimal, risky, and wasteful methods. At the end of the day, we as group members were fortunate to have been paired with engineers who could draw out these qualities from each of us and challenge us in areas where we lacked.

2. Prototype Manufacturing Processes and Candidates

Component #	Component Name	Process Candidates	Process Advantages	Process Disadvantages
1	Wheel Spoke	FDM 3D Printing	- Supports rapid prototyping and functional product creation - Facilitates manufacturing of intricate geometries	- Materials are more susceptible to breaking - Scaling up size poses challenges due to limited printer dimensions - Necessitates manual removal of support materials and unsmooth surface protrusions
		CNC Milling Machine	- Permits use of more robust, durable materials - Allows for manufacturing of complicated geometries - Higher Precision, desired tolerances more likely to be achieved	- Subtractive manufacturing of small facets may prove more challenging than additive methods - Increased material wastage
2	Wheel Hub	FDM 3D Printing	- Achieves satisfactory rates of rapid prototyping - Easy to manufacture replacement part in case of failure or innovation - Cost-effective regarding material use and waste	- Materials may have limited durability under real-world forces - Precision may vary depending on the printer used - Mass production could entail longer timeframes
		Injection Molding	- Enables utilization of a broader range of robust, durable materials - Potential for significantly faster production rates through automation	- Elevated material wastage due to early version failures - Potential for non-uniform material distribution during injection - Complexity arises from the necessity of numerous injection points

3	Handle	Universal Systems Laser Cutters	- Offers high precision with rapid production - Particularly efficient for simple extruded designs, resulting in faster production	- Incompatible with partial wood cuts - Partial charring of wood along cuts may occur, sacrificing precision - Limited by material thickness constraints
		ShopBot CNC Router	- Capable of high repeatability and precision - Ideal for mass production applications - Allows for different designs across depths	- May not be the most time-effective for single-product or simple-product scenarios - Also leaves unclean tabs that require manual cleaning
4	Wooden Wheel Supports	Universal Systems Laser Cutters	- Offers high precision with rapid turnaround - Particularly efficient for simple extruded designs, resulting in faster production.	- Incompatible with partial wood cuts - Partial charring of wood along cuts may occur - Limited by material thickness constraints
		ShopBot CNC Router	- Capable of high repeatability and precision - Ideal for mass production applications - Allows for different designs across depths	- May not be the most time-effective for single-product or simple-product scenarios - Also leaves unclean tabs that require manual cleaning
5	Bolts	Cold-forming	- Minimal material waste and energy use - Typically makes the bolts stronger through the process of work-hardening	- Size and pressure capabilities of the machine limits the size of the bolts - Many materials don't cold form very well; thus limited to a few ductile metals
		CNC Milling	- Incredibly high in accuracy, which is needed for such a small component - As it is CNC, it can be automated,	- Produces high amounts of material waste - May be time intensive for the sheer quantity needed in relation to the part

			partially saving time	
6	Nuts	Cold-forming	- Minimal material waste and energy use - High automatability and output - Work-hardening strengthens the nuts - High accuracy and precision within dimension	- Size and pressure capabilities of the machine limits the size of the bolts - Most materials don't cold form well, which limits metal choice
		Hot-forming	- Capable of making thicker and longer nuts - Size is much less limited than cold-forming - Process allows for more complicated geometries	- No work-hardening in hot forming - Requires post-processing in order to deal with surface scaling - Could potentially warp into the wrong shape - High energy cost through heat usage
7	Bearings	CNC Lathing	- High speed production - Saves a lot of time - Very hands-on manually; lots of control during the manufacturing process	- Requires a lot of machine maintenance - Limited in the shapes that can be produced - Subtractive manufacturing causes significant material wastage
		FDM 3D Printing	- Supports rapid prototyping and functional product creation - Facilitates manufacturing of intricate geometries	- Materials are more susceptible to breaking - Necessitates manual removal of support materials and unsmooth surface protrusions - Depending on printer quality, quality and resolution of print may not be optimal
8	Dowel	CNC Lathing	- High speed production - Saves a lot of time - Very hands-on manually; lots of control during the manufacturing	- Requires a lot of machine maintenance - Limited in the shapes that can be produced - Subtractive manufacturing causes significant material

			process	wastage
		Injection Molding	- Enables full control over size and shape of the final dowel - Highly repeatable - Compatible with a wide range of materials	- Can waste time and material creating molds for injection - Hard to iterate; creating multiple molds increases the amount of time it takes - Additive manufacturing can lead to lots of failed iterations until a satisfactory product is made
9	Spacers	CNC Milling Machine	- Permits use of more robust, durable materials - Allows for manufacturing of complicated geometries - Higher Precision, desired tolerances more likely to be achieved	- Subtractive manufacturing of small facets may prove more challenging than additive methods - Increased material wastage
		Injection Molding	- Enables full control over size and shape of the final spacers - Highly repeatable - Compatible with a wide range of materials	- Can waste time and material creating molds for injection - Hard to iterate; creating multiple molds increases the amount of time it takes - Additive manufacturing can lead to lots of failed iterations until a satisfactory product is made

Prototype Fabrication Procedure

We started by CAD-modeling and FDM-printing the wheel. This includes two wheel hubs and six wheel spokes, using the Creality Ender 3 printer. Subsequently, we assembled three wheel arms to each hub using M3 socket-head bolts and nuts. Wheel bearings were then press-fitted into the center of each hub. Moving to the wooden components, we utilized the Universal System laser cutter to precisely cut the wooden walker handles and wheel supports. Following this, dowels were cut to required lengths with a handsaw and inserted through the bearing-wheel configuration with an interference fit, along with spacers with clearance fits to create spacing between the wheels and wooden walker handles. Additionally, the dowels were passed through holes in the wooden walker arms, secured with wood glue to ensure firm attachment. Finally, the supports were glued onto similar faces within the wheel, allowing the wheels to roll at the same rate.

3. Tolerancing Table

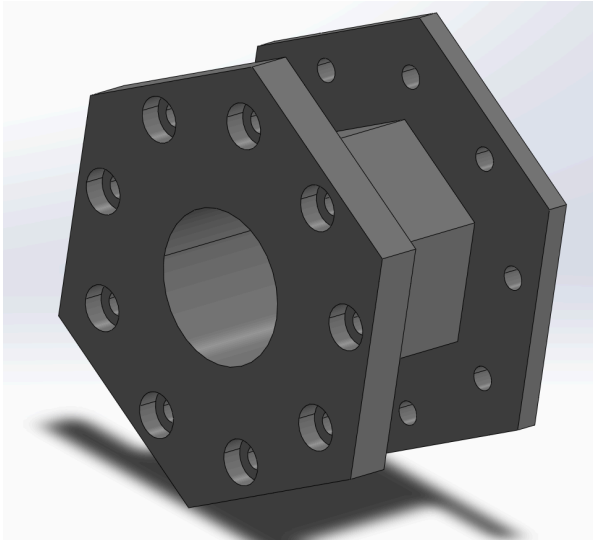
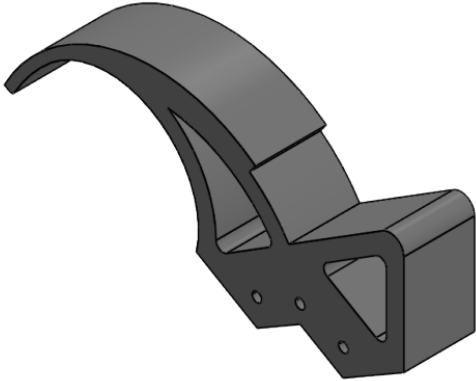
*Grades and tolerancing obtained from ANSI standards

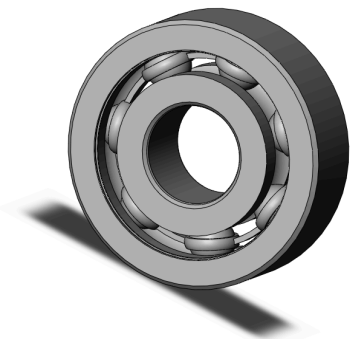
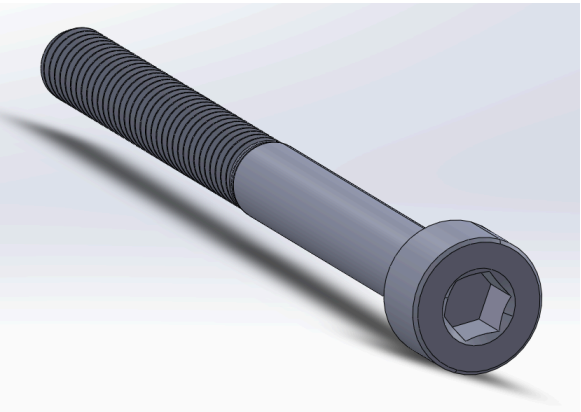
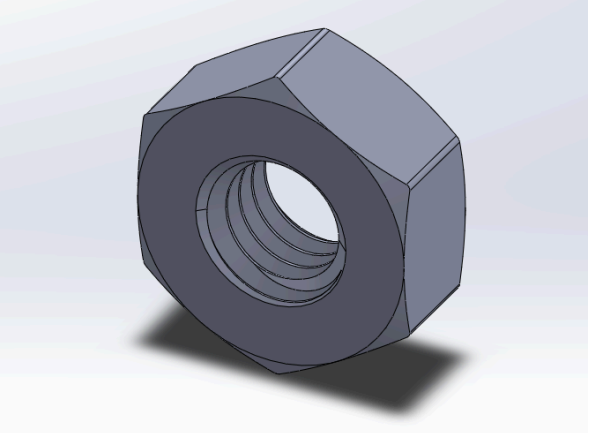
Fit #	Connects component #... (A)	...to component # or external object (B)	Function of fit	ANSI grade of fit	Component A critical dimension tolerance e.g. diameter 10.0±0.1 mm Tolerance given in thousandths of an in	Component B critical dimension tolerance Tolerance given in thousandths of an in	Can your chosen manufacturing process deliver the required tolerance for component A?	Can your chosen manufacturing process deliver the required tolerance for component B?
1	Bearing	Hub	To hold the bearing in place within the hub.	LN1	Shaft Bearing outer diameter: 0.866142in Tol: +1.0 to +0.5	Hole Hub hole diameter: 0.866142in Tol: +0.5 to 0	Yes. A CNC lathing procedure can achieve the desired tolerance when computer controlled	No. Very small tolerance to which the 3D printer used, could mimic the resolution needed.
2	Bearing	Dowel	To hold the Dowel within the inner housing in order to restrict axial horizontal sliding.	LN2	Hole Bearing inner diameter: 0.3125 in Tol: +0.6 to 0	Shaft Dowel diameter: 0.3125 in Tol: +1.0 to +0.6	Yes. A CNC lathing procedure can achieve the desired tolerance.	Yes. A CNC woodturning procedure can achieve the desired tolerance.
3	Bolt	Hub	To allow a connection between the spokes and the hub in order to hold them together. Also to allow for the bolt to physically be fit	RC4	Shaft M3 bolt: 0.1181in Tol: -0.3 to -0.7	Hole Hub bolt hole: 0.1181in Tol: +0.6 to 0	Yes. Cold-forming can achieve the required tolerance.	No. Very small tolerance to which the 3D printer used, could mimic the resolution needed.

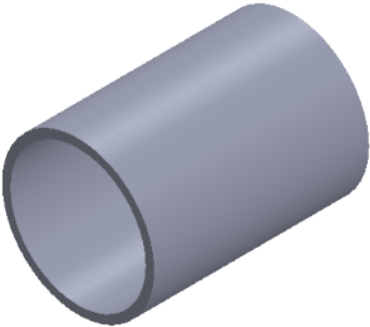
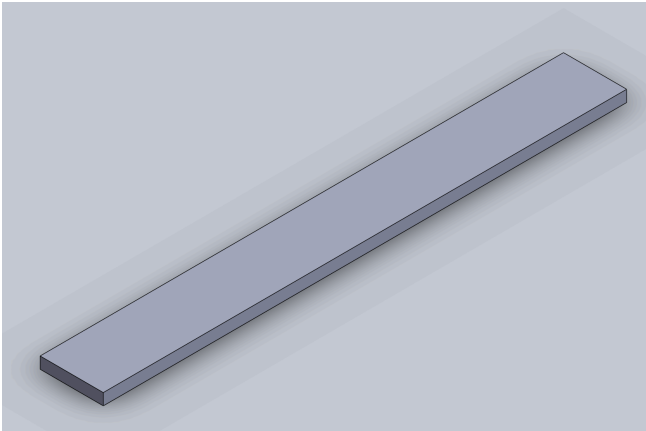
			throughout the hub.					
4	Bolt	Hub Counterbore	To allow for the bolt to fit flush within the hub	RC8	Shaft M3 bolt: 0.1181in Tol: -2.5 to -3.5	Hole Counterbore diameter: 0.2559 Tol: +2.2 to 0	Yes. Cold-forming can achieve the required tolerance.	No. Very small tolerance to which the 3D printer used, could mimic the resolution needed.
5	Handle	Dowel	To restrict the dowels axial movement within the handle, and well as the radial rotation within the handle	LN2	Hole Handle hole: 0.3125 in Tol: +0.6 to 0	Shaft Dowel diameter: 0.3125 in Tol: +1.0 to +0.6	Yes. Laser Cutting can achieve the required tolerance.	Yes. A CNC woodturning procedure can achieve the desired tolerance.
6	Spoke	Hub	To provide rigidity within the wheel during rotation.	LN1	Shaft Spoke width: 1 in Tol: +0.5 to 0	Hole Hub gap: 1 in Tol: +1.0 to +0.5	No. Very small tolerance to which the 3D printer used, could mimic the resolution needed.	No. Very small tolerance to which the 3D printer used, could mimic the resolution needed.
7	Spacer	Dowel	To provide a fixed distance between the wheel and the handle, while also allowing for free rotation of the Dowel.	RC2	Hole Spacer inner diameter: 0.3125 in Tol: +0.4 to 0	Shaft Dowel diameter: 0.3125 in Tol: -0.2 to -0.45	Yes, CNC alone has very high precision. The combination with lathing for the dowel also favors precision, allowing	Yes. A woodturning procedure can achieve the desired tolerance.

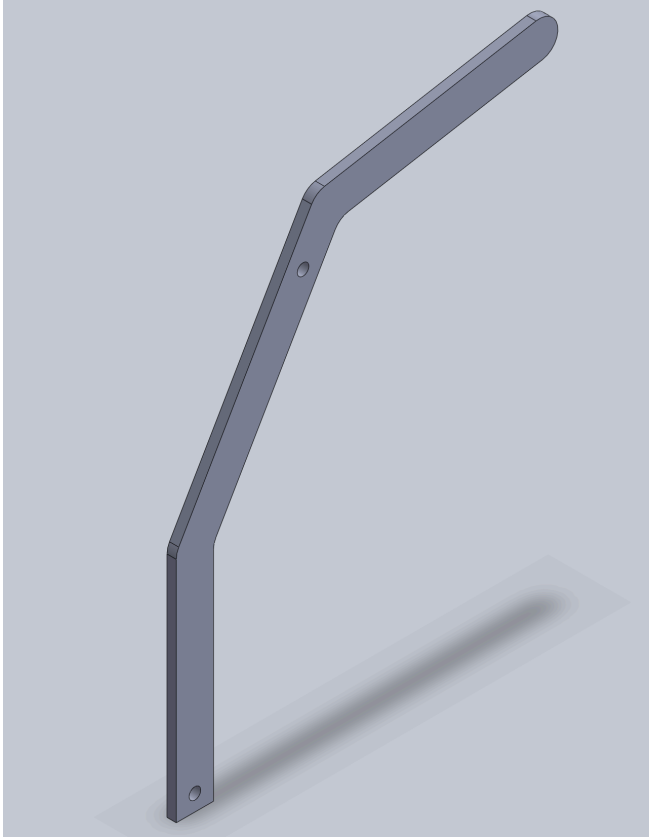
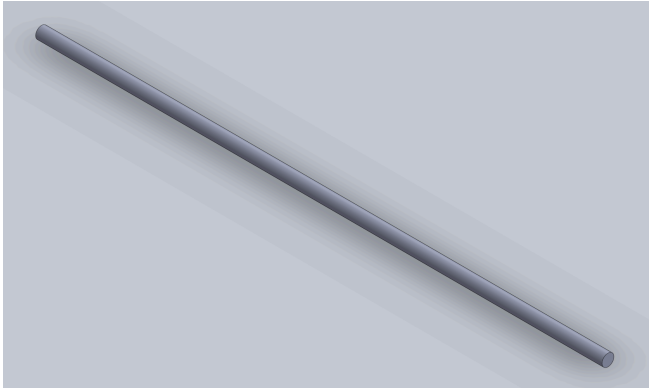
							for parts to be within tolerance	
8	Bolt	Spoke	To allow for the bolt to physically be fit throughout both the spoke and hub.	RC4	Shaft M3 bolt: 0.1181in Tol: -0.3 to -0.7	Hole Spoke bolt hole: 0.1181in Tol: +0.6 to 0	Yes, cold-forming of the bolt provides consistent production, falling under tolerance if properly set up.	No. Very small tolerance to which the 3D printer used, could mimic the resolution needed.
9	Nut	Bolt	Secure all wheel components together	Thread Fit	N/A	N/A	Yes, cold-forming of the bolt provides consistent production, falling under tolerance if properly set up.	Yes, cold-forming of the bolt provides consistent production, falling under tolerance if properly set up

4. Components + Images

Component Number	Component	Quantity	Picture (Rendered CAD component is fine)
1	Wheel Hub	2	 The wheel hub is the central component used for creating a mate with the spokes.
2	Wheel Spoke	6	 The spoke is the part of the wheel in which leverages on the stair, and rolls. It sits within the hub.

3	Bearing	4	 The bearing is what is fitted into the hub hole, and slid through the axle, allowing for free rotation of the wheel.
4	Bolt	18	 The bolt is used to align and hold the spokes within the hub.
5	Nut	18	 The nut is used to fasten the bolt, securing the wheel structure.

6	Spacer	2	 The spacer allows for a fixed distance between the wheel and the handle, in order for the handle to not collide with the overhanging bolts.
7	Wheel Support	3	 The wheel support is glued to similar faces of both wheels in order to allow for simultaneous rotation.

8	Handle	2	 The handle is where the axle sits. It provides a grip for the user.
9	Dowel	2	 This is the axle in which the bearings, as well as the handles slide onto. The second instance within the handle is to provide rigidity.

5. At-Scale Manufacturing Processes and Candidates

Component #	Component Name	Process Candidates	Process Advantages	Process Disadvantages
1	Wheel Spoke (Aluminum)	Sand-casting	- Production process is very simple once you have the mold - Molds can be complex yet inexpensive - Production itself is relatively inexpensive - Casting allows for a wide material choice	- Requires manual processing, which leads to variability in product - Product may have low strength and dimensional accuracy - Surface quality is not smooth and may not be consistent
		CNC Milling Machine	- Permits use of more robust, durable materials - Maintains comparable levels of automatability - Extremely precise, fast, and efficient compared to sand-casting	- Subtractive manufacturing of small facets may prove more challenging than additive methods - Increased material wastage - Requires a high level of skill to operate a CNC machine
2	Wheel Hub (Aluminum)	Sand-casting	- Production process is very simple once you have the mold - Molds can be complex yet inexpensive - Production itself is relatively inexpensive - Casting allows for a wide material choice	- Requires manual processing, which leads to variability in product - Product may have low strength and dimensional accuracy - Surface quality is not smooth and may not be consistent
		CNC Milling Machine	- Permits use of more robust, durable materials - Maintains comparable levels of automatability - Extremely precise, fast, and efficient compared to sand-casting	- Subtractive manufacturing of small facets may prove more challenging than additive methods - Increased material wastage - Requires a high level of skill to operate a CNC machine

3	Handles (Aluminum alloy or brass tubes)	CNC Lathing	- High speed production - Saves a lot of time - Very hands-on; lots of control during the manufacturing process	- Requires manual labor and high skill interaction - Requires a lot of machine maintenance - Limited in the shapes that can be produced - Subtractive manufacturing causes significant material wastage
		Tube drawing	- Very economical in material/money use - Has the added strength of cold-working - Highly repeatable with precise dimensions	- Diminishing surface finish with thicker tubes - Requires a drawing machine and stock/die that are the right size and shape to work with the machine - Relatively slower process
4	Wheel Supports (Aluminum or wood)	Universal Systems Laser Cutters	- Very high precision and high speed - Highly repeatable due to computational automatability with G-code programming - Works with a variety of materials	- Potentially harmful gases and fumes from the process - Limited on material thickness as laser needs to be able to cut through the material - High energy consumption - May cost too much for the simplicity of the task at hand
		Vertical bandsaw	- Very simple, low-effort cutting process for simple shapes - High efficiency and moderate precision for simple cuts - Wider variety of materials and thicknesses available	- Requires manual operation, which introduces possible human error - Wastes material based on the size of stock material - Doesn't leave the cleanest finish on the edges; may cause chips
5	Bolts	Cold-forming	- Minimal material waste and energy use - Typically makes the bolts stronger through the process of work-hardening	- Size and pressure capabilities of the machine limits the size of the bolts - Many materials don't cold form very well; thus limited to a few ductile metals
		CNC Milling	- Permits use of more robust, durable	- Subtractive manufacturing of small facets may prove

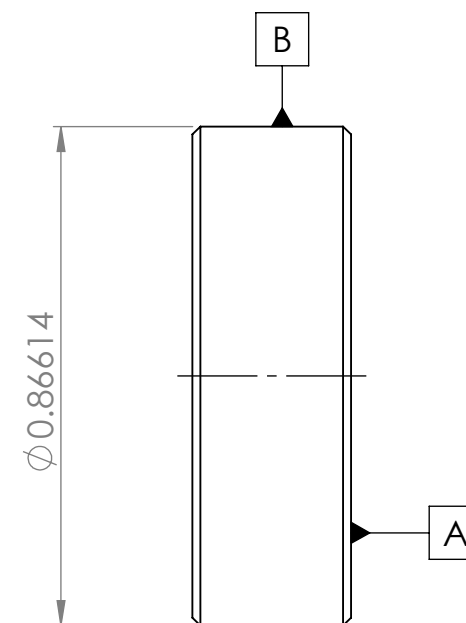
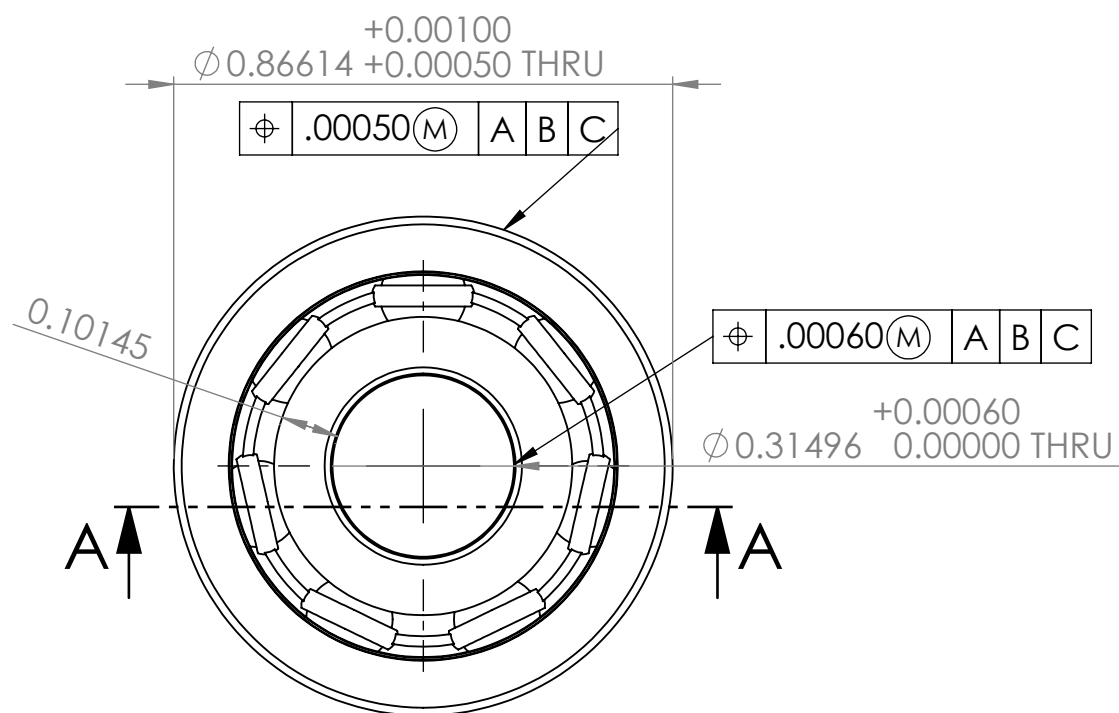
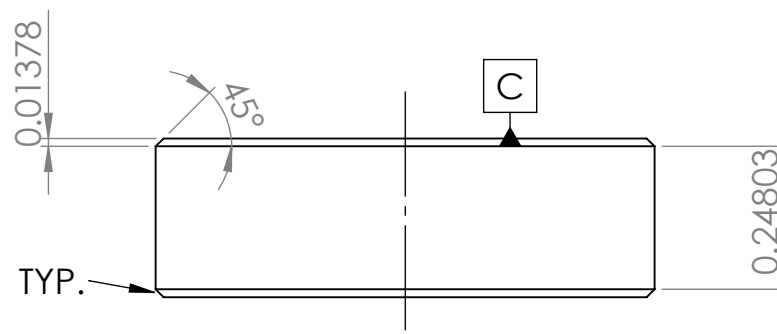
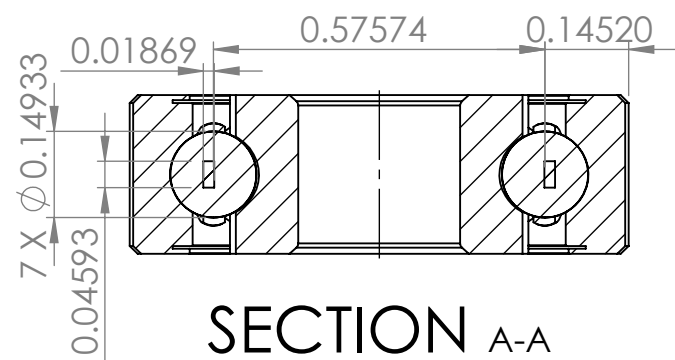
			materials - Maintains comparable levels of automatability - Extremely precise, fast, and efficient	more challenging than additive methods - Increased material wastage - Requires a high level of skill to operate a CNC machine
6	Nuts	Cold-forming	- Minimal material waste and energy use - High automatability and output - Work-hardening strengthens the nuts - High accuracy and precision within dimension	- Size and pressure capabilities of the machine limits the size of the bolts - Most materials don't cold form well, which limits metal choice
		Hot-forming	- Capable of making thicker and longer nuts - Size is much less limited than cold-forming - Process allows for more complicated geometries	- No work-hardening in hot forming - Requires post-processing in order to deal with surface scaling - Could potentially warp into the wrong shape - High energy cost through heat usage
7	Bearings	CNC Lathing	- High speed production - Saves a lot of time - Very hands-on; lots of control during the manufacturing process	- Requires manual labor and high skill interaction - Requires a lot of machine maintenance - Limited in the shapes that can be produced - Subtractive manufacturing causes significant material wastage
		FDM 3D Printing	- Supports rapid prototyping and functional product creation - Facilitates manufacturing of intricate geometries	- Materials are more susceptible to breaking - Necessitates manual removal of support materials and unsmooth surface protrusions - Depending on printer quality, quality and resolution of print may not be optimal
8	Dowel (Aluminum)	CNC Lathing	- High speed production	- Requires manual labor and high skill interaction

			- Saves a lot of time - Very hands-on; lots of control during the manufacturing process	- Requires a lot of machine maintenance - Limited in the shapes that can be produced - Subtractive manufacturing causes significant material wastage
		Tube drawing	- Very economical in material/money use - Has the added strength of cold-working - Highly repeatable with precise dimensions	- Diminishing surface finish with thicker tubes - Requires a drawing machine and stock/die that are the right size and shape to work with the machine - Relatively slower process
9	Spacers	CNC Lathing	- High speed production - Saves a lot of time - Very hands-on; lots of control during the manufacturing process	- Requires manual labor and high skill interaction - Requires a lot of machine maintenance - Limited in the shapes that can be produced - Subtractive manufacturing causes significant material wastage
		Injection Molding	- Enables full control over size and shape of the final spacers - Highly repeatable - Compatible with a wide range of materials	- Can waste time and material creating molds for injection - Hard to iterate; creating multiple molds increases the amount of time it takes - Additive manufacturing can lead to lots of failed iterations until a satisfactory product is made

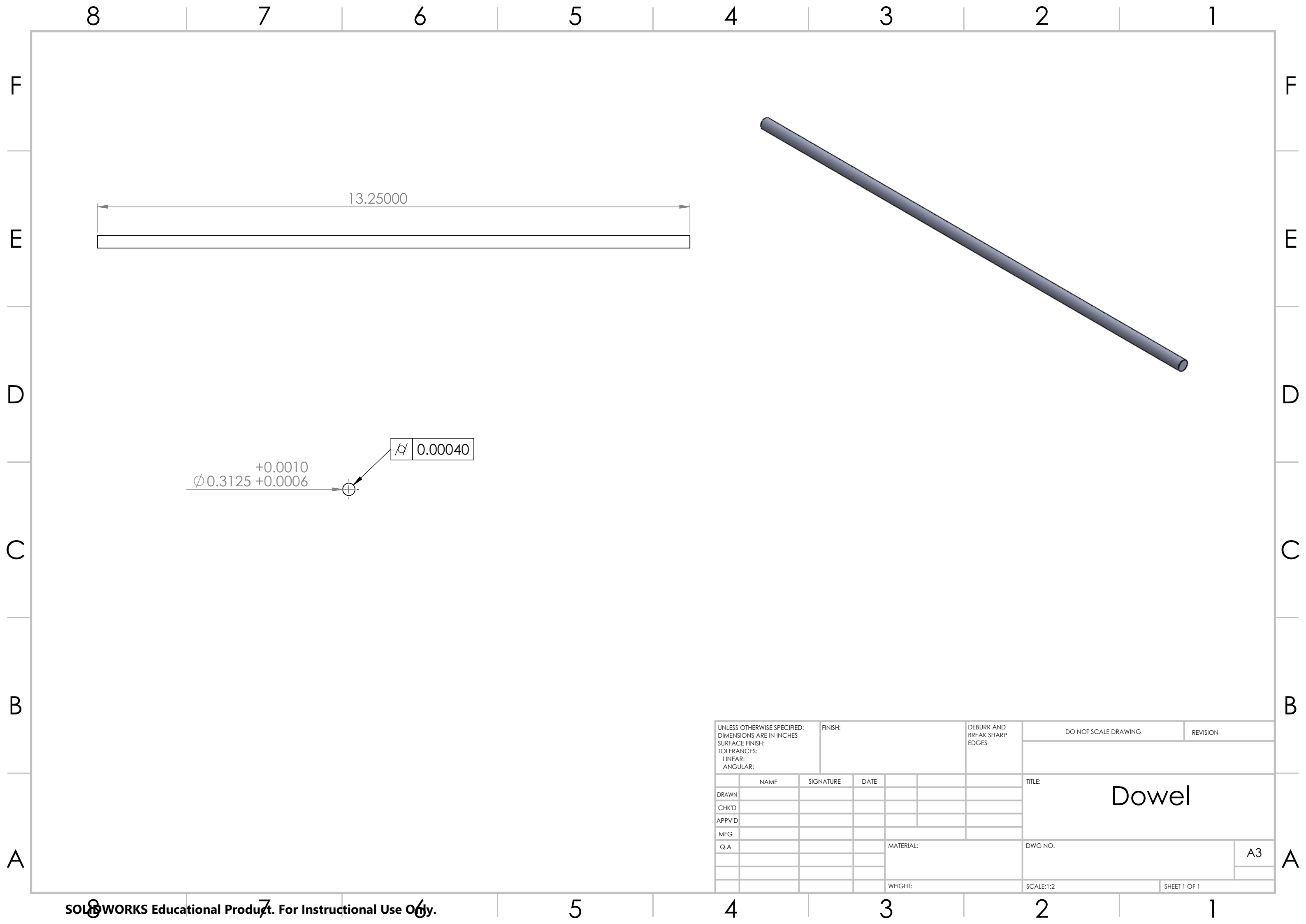
At-Scale Fabrication Procedure

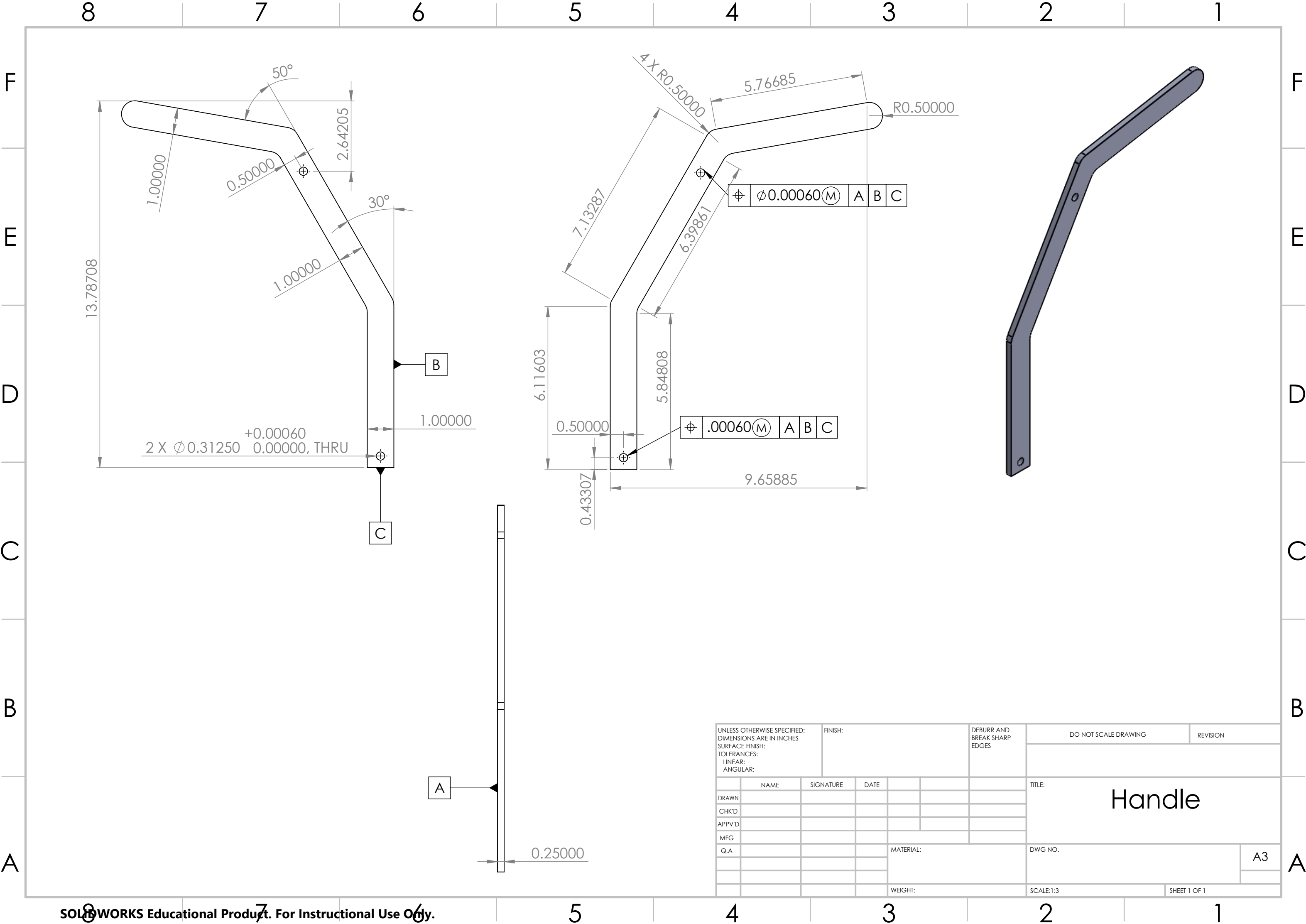
To manufacture the walker, we begin by CNC milling the wheel spokes and hubs from aluminum stock. Our main hardware for our walker are bolts and nuts which will both be mass produced through cold forming. The three wheel spokes are then assembled to each hub using the socket-head bolts and nuts. The bearings are created through a combined process of tube drawing (cylindrical shells) and stamping (inner balls). Next, we press-fit the bearings by an interference fit at the center of each hub. Our walker handles are created by being lathe-turned from aluminum and welded together to form the walker's structural frame. The wheel supports are manufactured from aluminum by being cut by a vertical bandsaw. They are then welded onto the wheel spokes to ensure equal turning of both wheels. Additionally, aluminum dowels are manufactured using a lathe, and after drilling holes into the handles with a drill press, the dowels are then fitted through these holes with an interference fit, and then welded for extra security. Our spacers are manufactured through a process of tube drawing, then being cut by a fablight to obtain the proper length. Then, the dowels make a clearance-fit with these spacers. This creates a necessary gap between the wheel and the handle, allowing for free rotation. Finally, to secure the dowels in place, they are welded to the handles, ensuring the structural integrity of the walker.

Drawings on Following Pages:

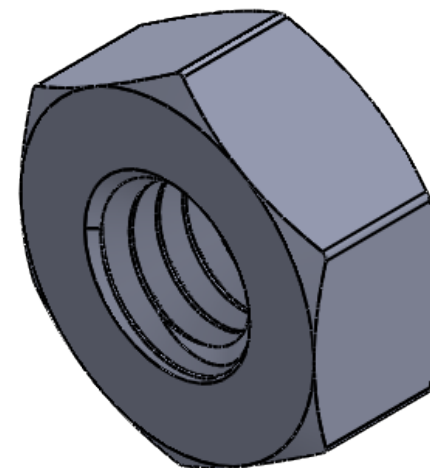
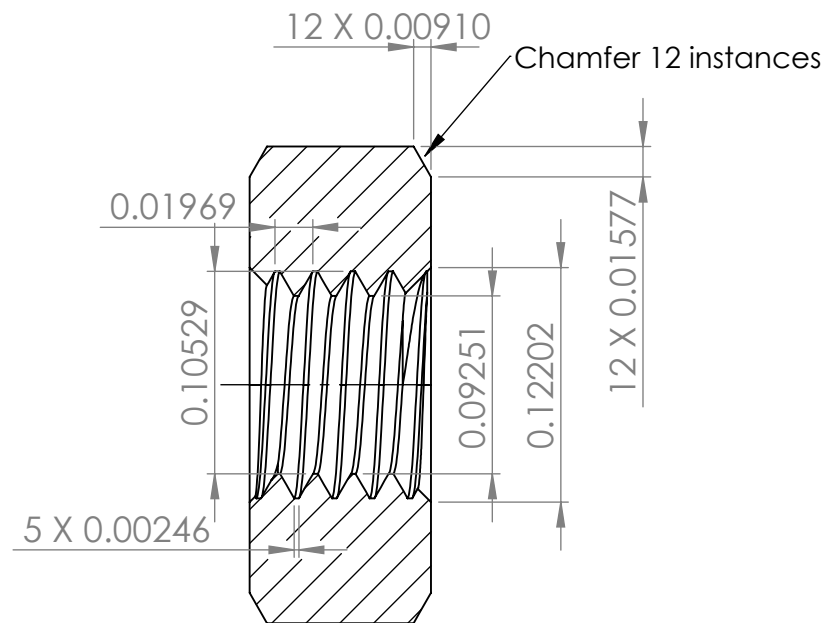
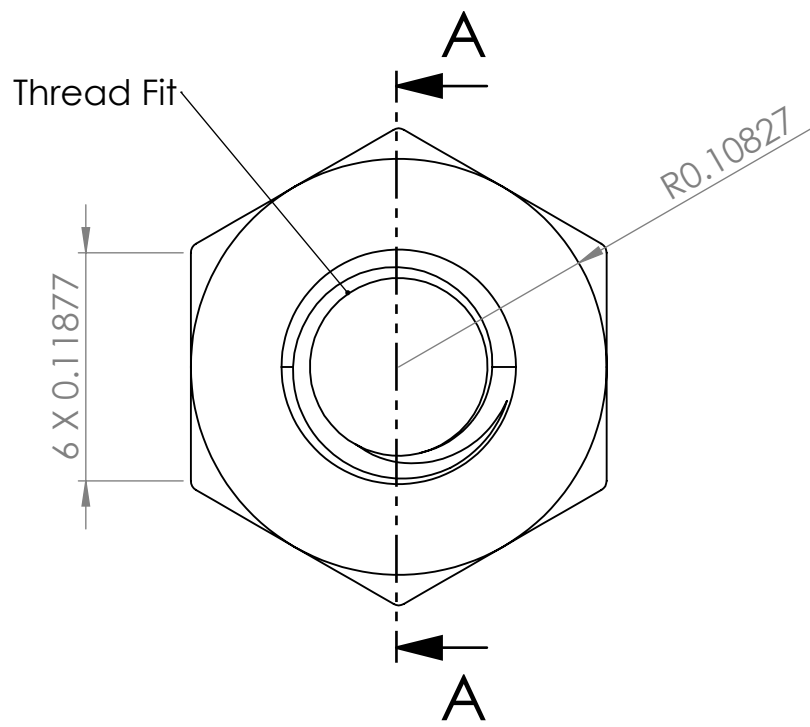


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DRAWN				NAME		SIGNATURE		DATE		TITLE:	
CHK'D										Ball Bearing	
APPV'D											
MFG											
Q.A											
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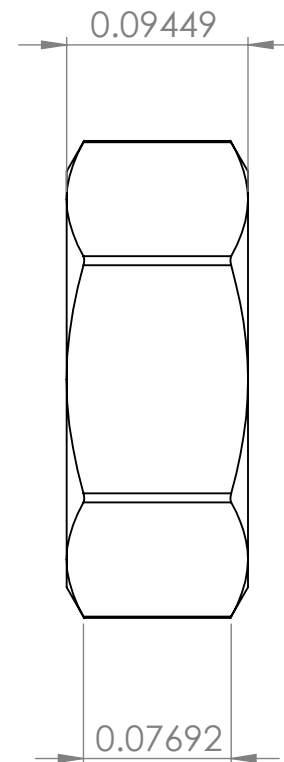
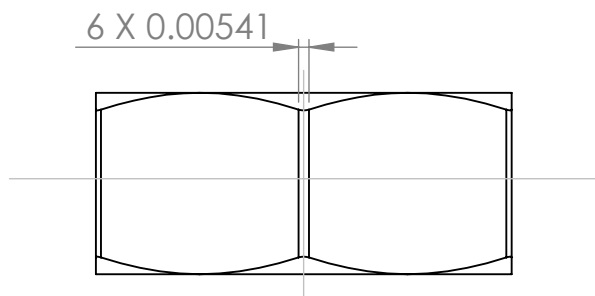




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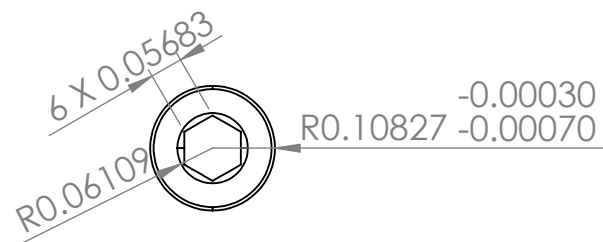


SECTION A-A

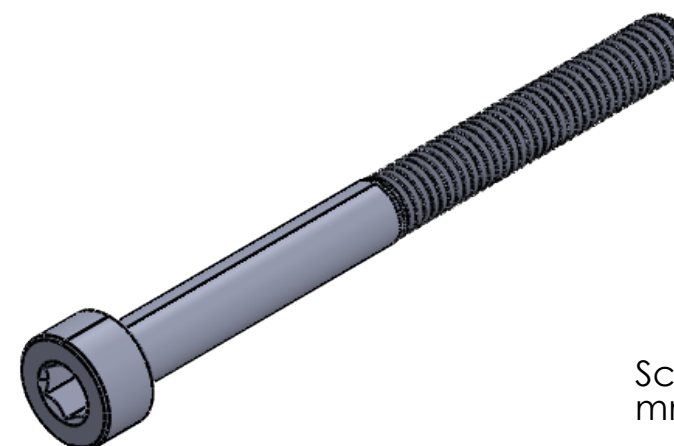


Note:
Nut is based on M3 X
0.5 mm thread
dimensioning.

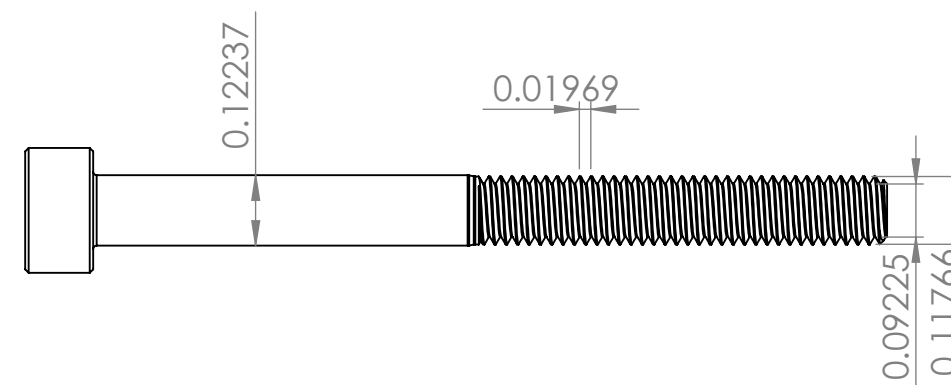
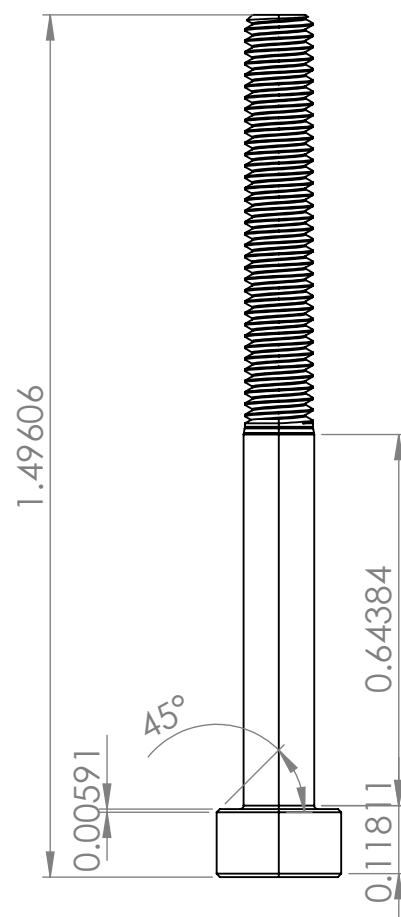
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DRAWN				NAME		SIGNATURE		DATE		TITLE: Nut	
CHK'D											
APPV'D											
MFG											
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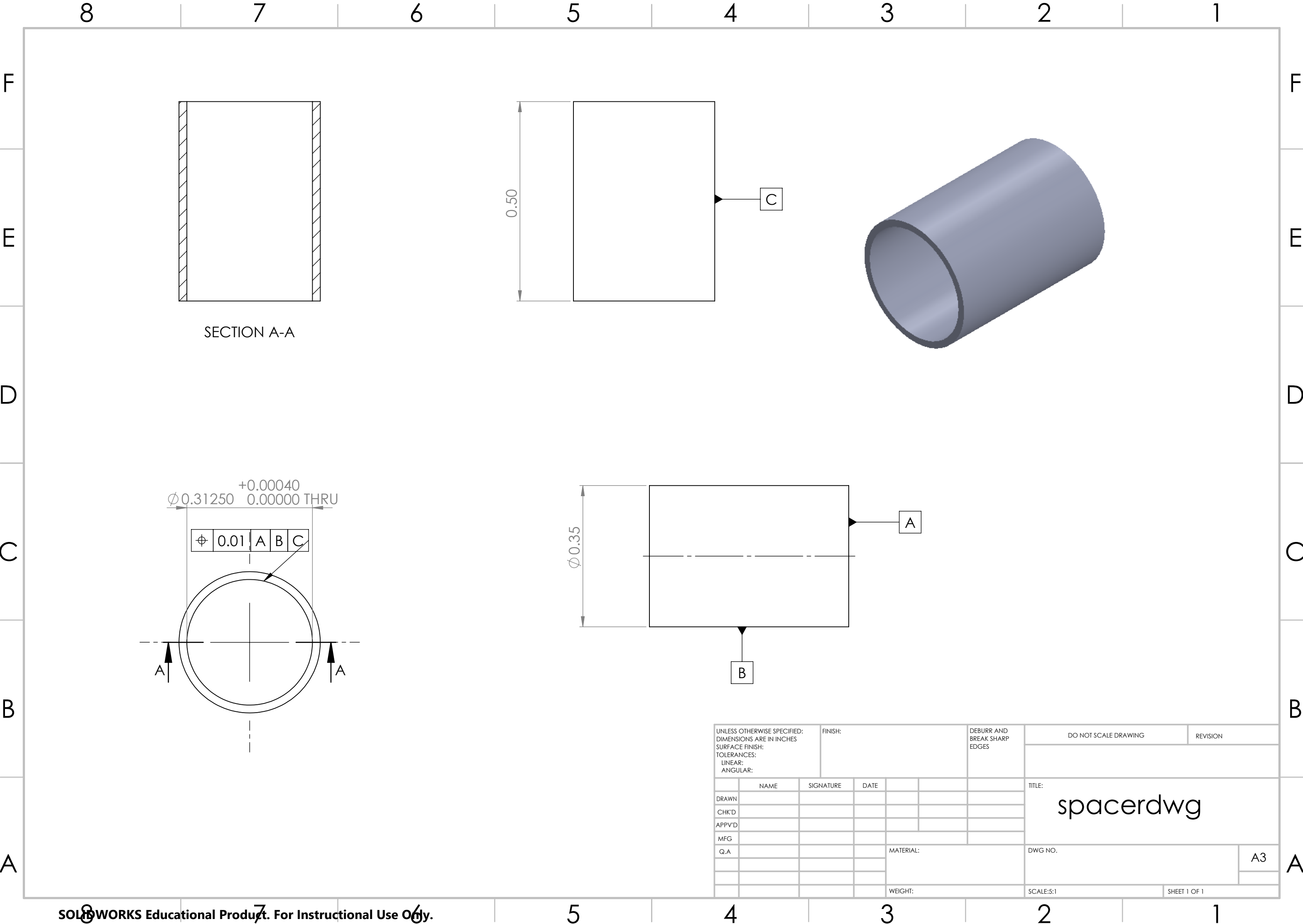
Thread Fit



ScREW is based on M3 X 0.5 mm dimensioning



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DRAWN																
CHK'D																
APPV'D																
MFG																
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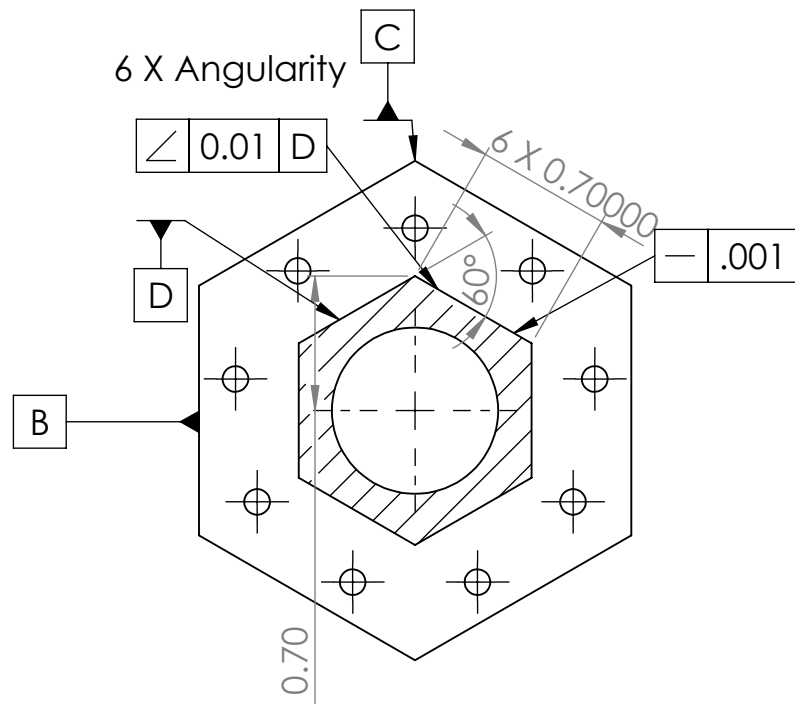
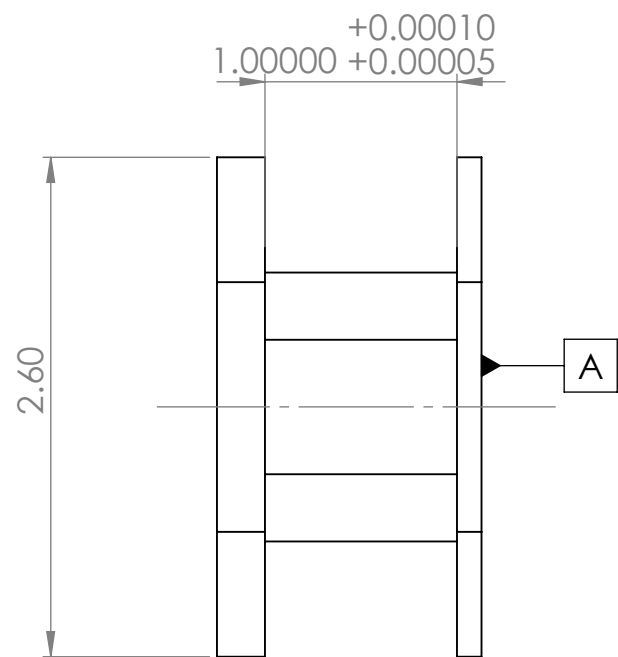
E

D

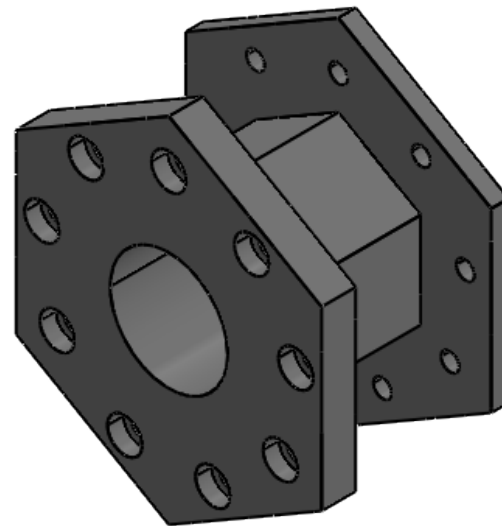
C

B

A



SECTION A-A



F

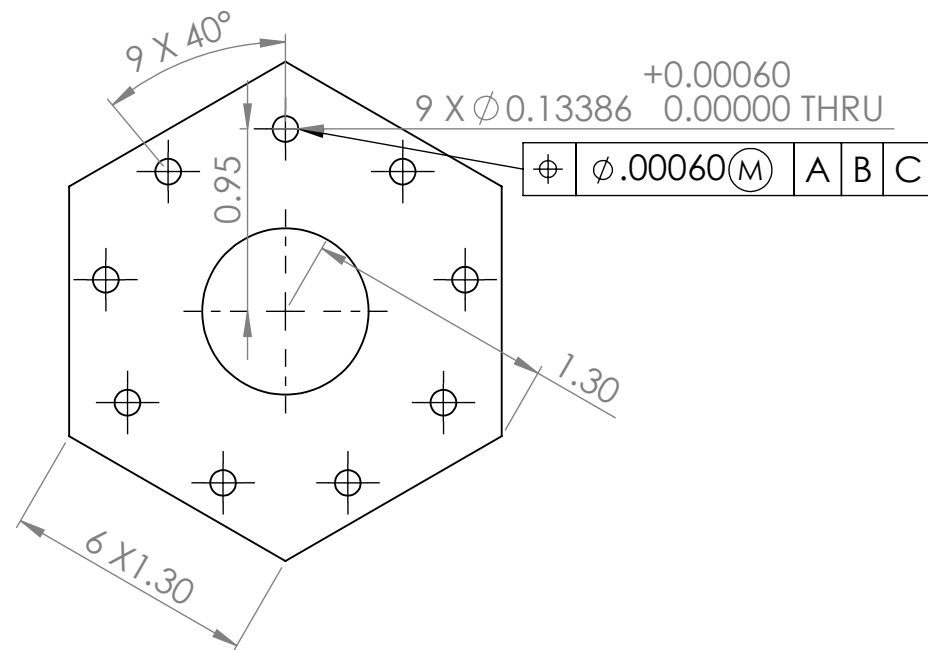
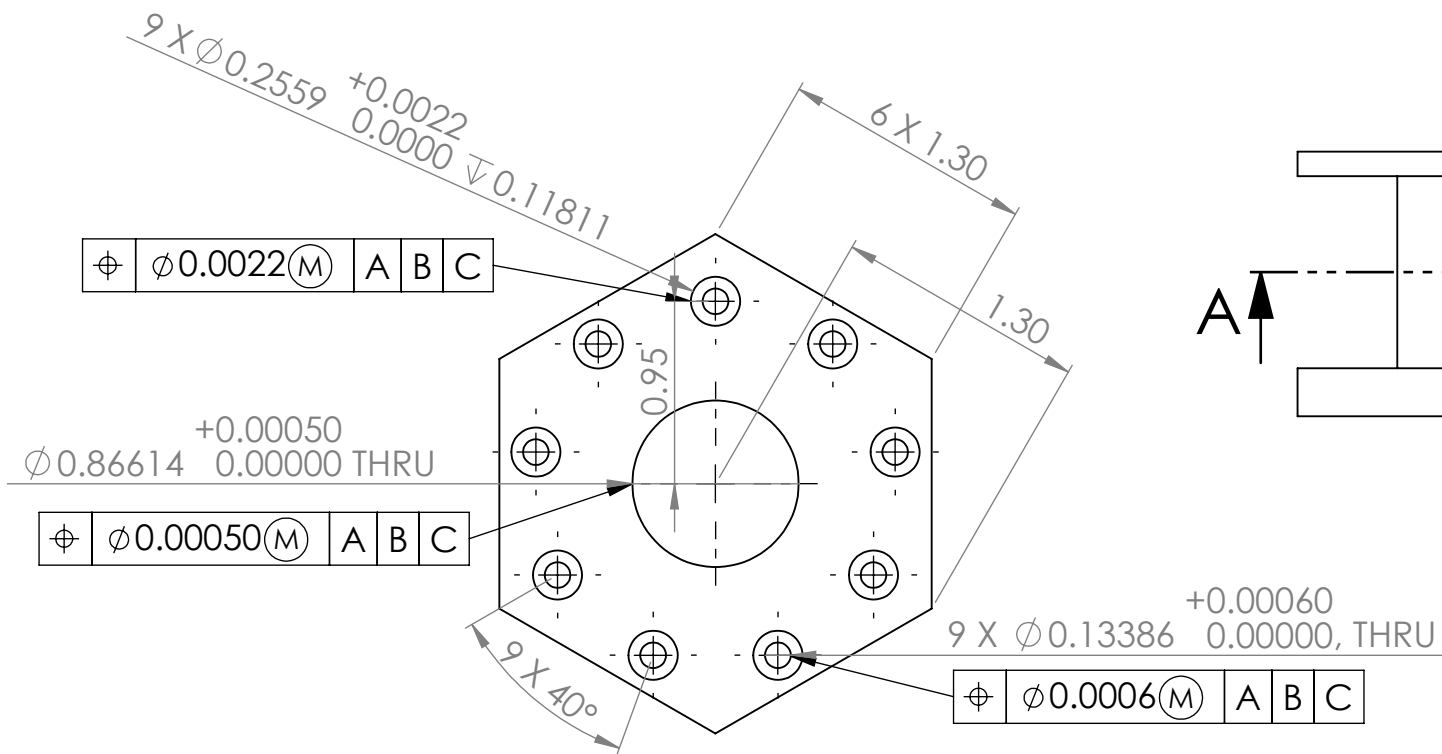
E

D

C

B

A



8 7 6 5 4 3 2 1

F

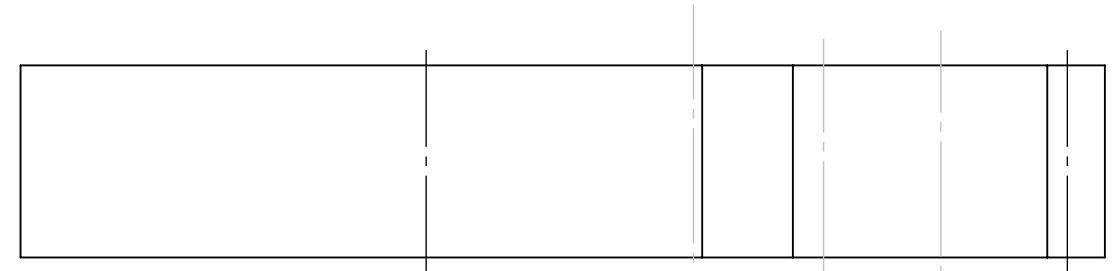
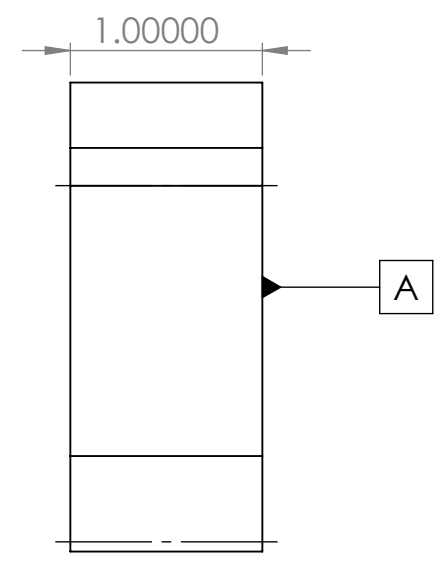
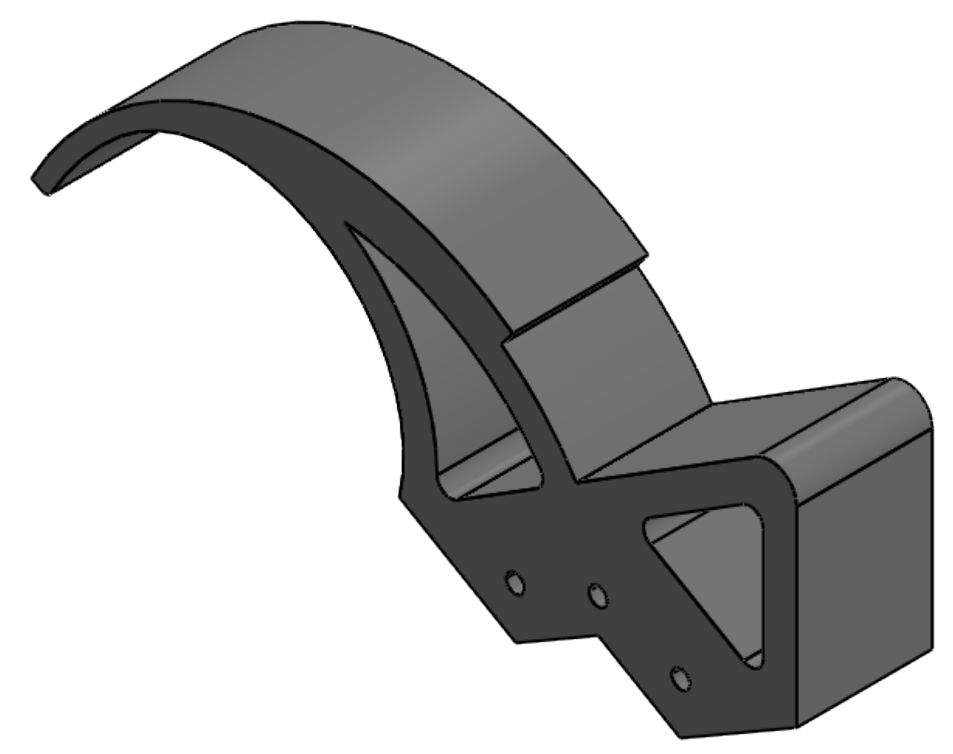
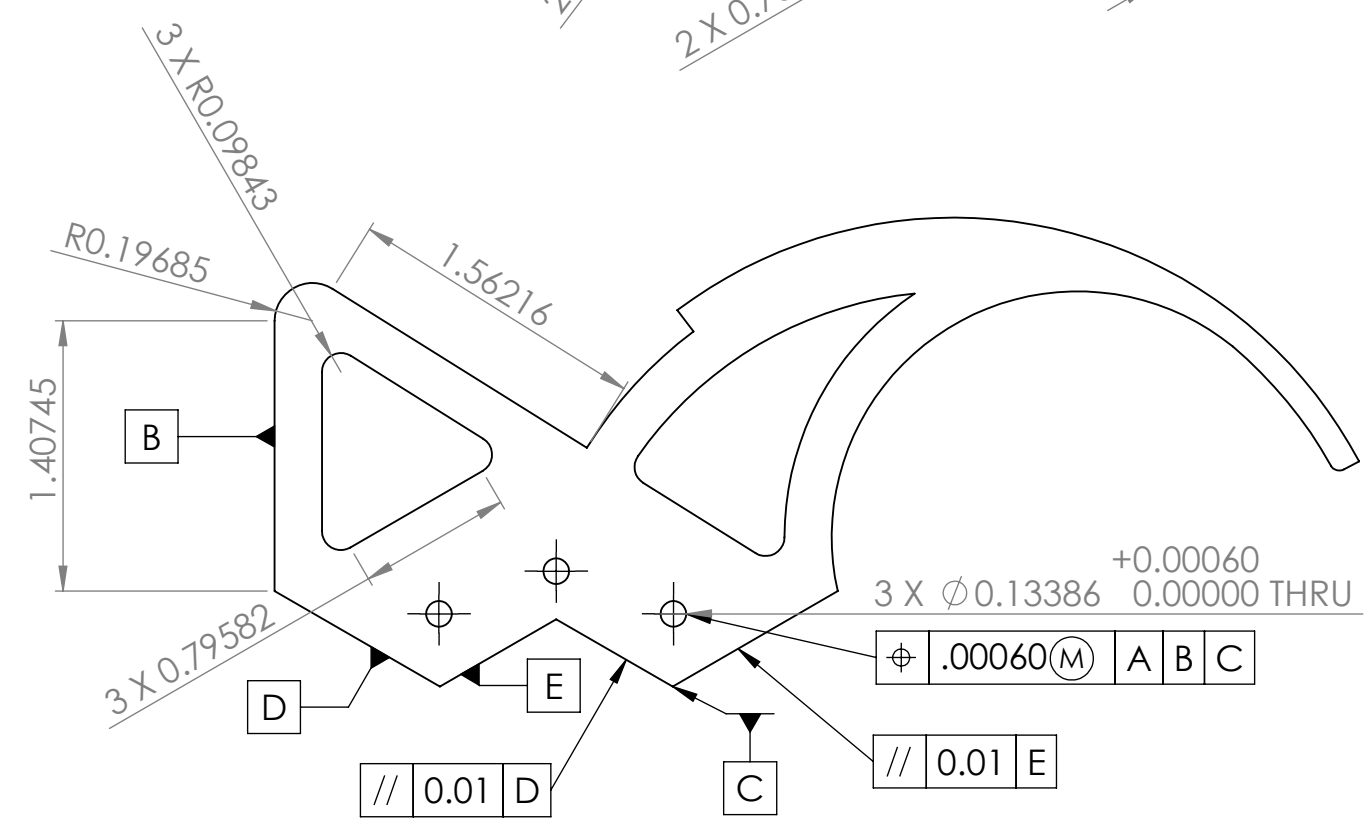
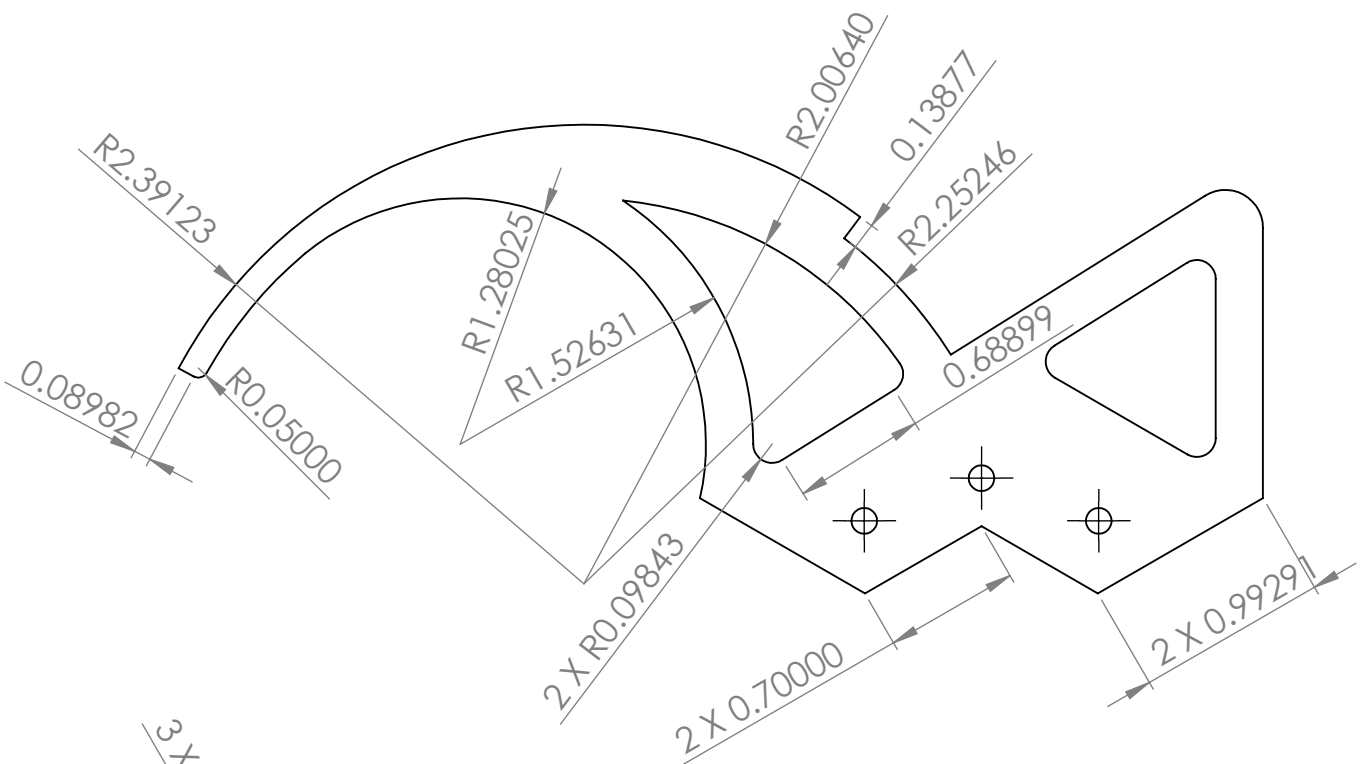
E

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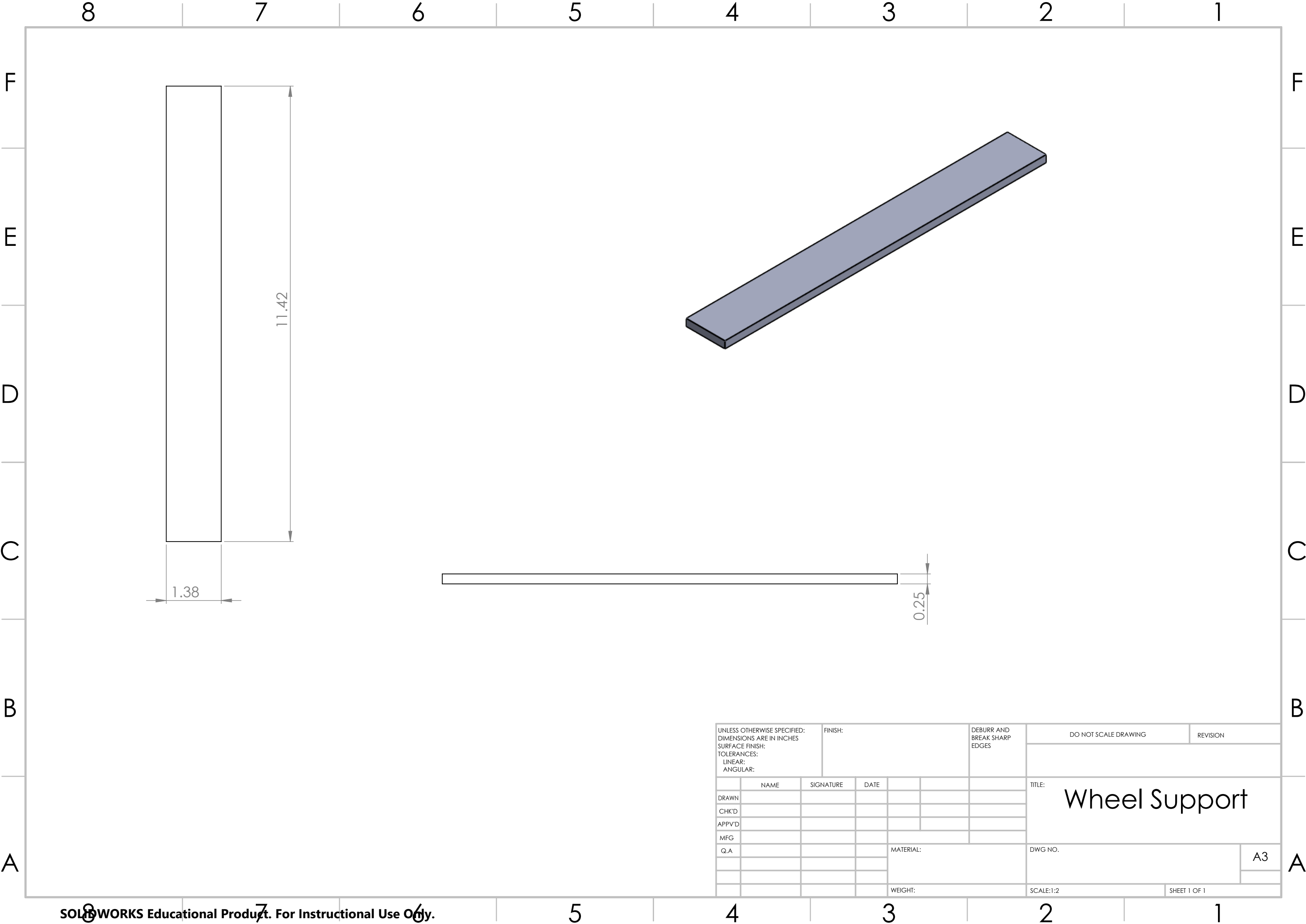
C

B

A



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:				FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION	
DRAWN				NAME		SIGNATURE		DATE		TITLE: Wheel Spoke	
CHK'D											
APPV'D											
MFG											
Q.A								MATERIAL:		DWG NO.	
										A3	
								WEIGHT:		SCALE:1:1	
										SHEET 1 OF 1	



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:				FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING		REVISION	
DRAWN		NAME		SIGNATURE		DATE		TITLE:		Wheel Support	
CHK'D											
APPV'D											
MFG											
Q.A											
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								WEIGHT:		SCALE:1:2	
										SHEET 1 OF 1	

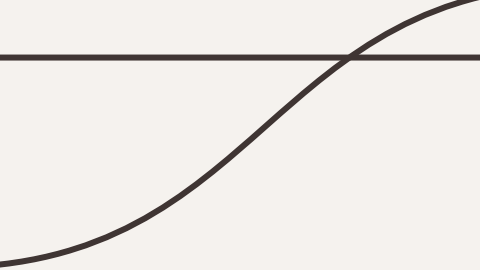
8. Elevator Pitch + Transcript

Recorded Pitch

https://drive.google.com/file/d/1G1yRUEv4QJf5EtRD6WPoPIUPzLgn6xOu/view?usp=drive_link

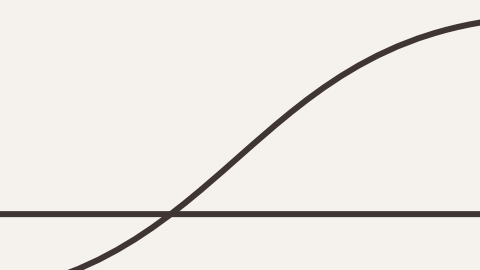
Transcript

With roughly 4 million elders needing mobility aid devices daily, current walkers force elders to endure a slow, uncomfortable and tiring process to go up stairs. Users must hold the walker in one hand and the handrail in the other, all to step up one leg at a time. Our group's design eliminates this inconvenient procedure by developing a revolutionary walker wheel design that allows users to ascend stairs smoothly without needing to reposition their walker. Our innovative mechanism maintains the walker at surface level when the user climbs stairs. As each wheel spoke rotates over the step, the user can independently and comfortably go up stairs needing only to apply force horizontally. This means slightly leaning into the walker in order for the wheel to roll over the stair. This design ensures users remain comfortable in any instance of a step. Unlike our competitors, we designed our walker to empower users to independently navigate stairways, minimizing time and improving accessibility.



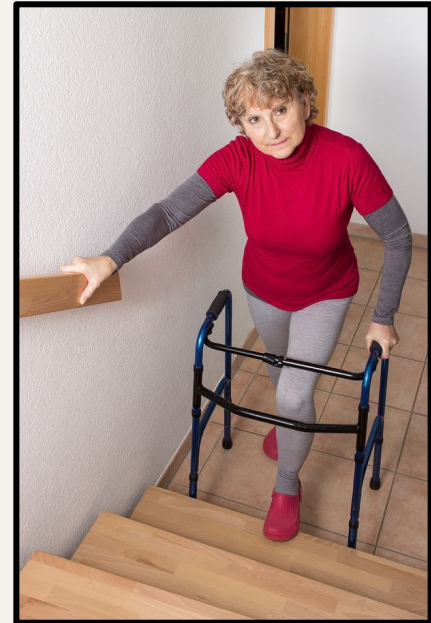
Revolutionary Walker Wheel Design

Group 6 - Rafael Petrosian, Gabriel Sianez, Joshua Oh, and
Catherine Harada



Problem

Elderly people with walkers have difficulty going up the stairs.



Solution

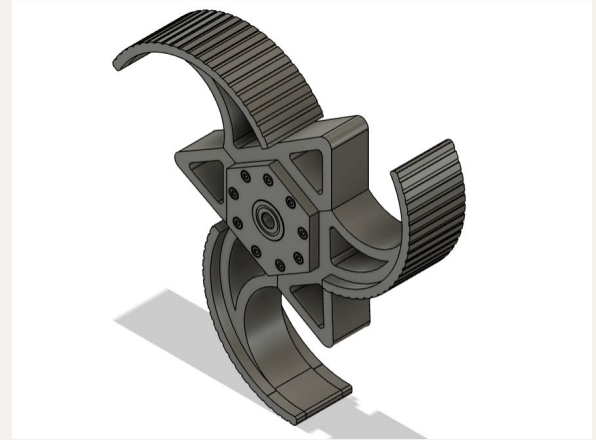
An innovative mobility aid that replaces walkers go up stairs easily



Our Team's Design Idea

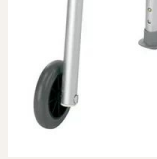
A wheel designed for stairs

We designed a **curved-spoke tri wheel walker** that rolls up stairs without needing to reposition the walker.

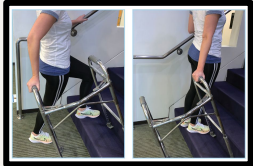


[1] Y. Kim, J. Kim, H. Kim, and T. Seo, Curved-Spoke Tri-Wheel Mechanism for Fast Stair-Climbing, [https://repository.hanyang.ac.kr/bitstream/20.500.11754/158828/1/Curved-Spoke %20Tri-Wheel%20Mechanism%20for%20Fast%20Stair-Climbing.pdf](https://repository.hanyang.ac.kr/bitstream/20.500.11754/158828/1/Curved-Spoke%20Tri-Wheel%20Mechanism%20for%20Fast%20Stair-Climbing.pdf) (accessed May 2, 2024).

Differentiation



	One-Wheel System	Tri-Wheel System	Our Solution
Roll Up Stairs	X	✓	✓
Minimize Collisions	X	✓	✓
No Sideways Repositioning Required	X	✓	✓
Travel at Different Heights	X	X	✓



Implications

4,094,000 senior citizens in the United States use walkers.

60% of users reported problems with their walking frames, classifying it as difficult to use.

Our wheel design presents them with...

1. Freedom of Mobility
2. Independence in navigating stairs



Works Cited

- [1] Y. Kim, J. Kim, H. Kim, and T. Seo, Curved-Spoke Tri-Wheel Mechanism for Fast Stair-Climbing, <https://repository.hanyang.ac.kr/bitstream/20.500.11754/158828/1/Curved-Spoke%20Tri-Wheel%20Mechanism%20for%20Fast%20Stair-Climbing.pdf> (accessed May 2, 2024).
- [2] N. M. Gell, R. B. Wallace, A. Z. LaCroix, T. M. Mroz, and K. V. Patel, “Mobility Device Use in Older Adults and Incidence of Falls and Worry About Falling: Findings from the 2011–2012 National Health and Aging Trends Study,” *Journal of the American Geriatrics Society*, vol. 63, no. 5, pp. 853–859, May 2015, doi: <https://doi.org/10.1111/jgs.13393> (accessed May 2, 2024).
- [3] Sibylle Brunhilde Thies, S. Bevan, M. Wassall, and D. Howard, “Evaluation of a novel biomechanics-informed walking frame, developed through a Knowledge Transfer...,” ResearchGate, Nov. 13, 2023. https://www.researchgate.net/publication/375608127_Evaluation_of_a_novel_biomechanics-informed_walking_frame_developed_through_a_Knowledge_Transfer_Partnership_between_biomechanists_and_design_engineers (accessed May 02, 2024).